Qatar University

Department of Mathematics, Statistics and Physics

Department of Mathematics and Statistics

Math 231 – Linear Algebra Quiz 3

Sections 4.1–4.4 (Real Vector Spaces, Subspaces, Spanning Sets, Linear Independence)

Fall 2026 • Thursday, 5 November 2026 • Time: 20 minutes • Total: 10 marks

Name: _____ ID: _____ Section: _____ **Version A**

Tick exactly one box for each True/False statement and exactly one option for each multiple-choice question; an item with more than one answer receives zero. Show all your work in Problem 3. Programmable calculators and mobile phones are not allowed.

Problem 1. True or False [2 marks]

Each statement is worth 0.5 mark. Tick **True** if the statement is always true, otherwise tick **False**. No justification is needed.

	True	False
(a) In any vector space V, $0\mathbf{u} = \mathbf{0}$ for every vector $\mathbf{u}$ in V.	■	□
(b) The set of all invertible 2×2 matrices is a subspace of M_{22}.	□	■
(c) If $S = \{\mathbf{w}_1, \ldots, \mathbf{w}_r\}$ is a set of vectors in a vector space V, then $\text{span}(S)$ is a subspace of V.	■	□
(d) If $\{\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3\}$ is linearly dependent, then $\mathbf{v}_3$ must be a linear combination of $\mathbf{v}_1$ and $\mathbf{v}_2$.	□	■

True. This is Theorem 4.1.1(a).

False. The zero matrix is not invertible, so condition (a) of the Subspace Test fails (Example 4.2.4).

True. Theorem 4.3.1.

False. The Remark after Theorem 4.4.1 says *at least one* vector is a linear combination of the others, not necessarily $\mathbf{v}_3$: for $\mathbf{v}_1 = (1,0,0)$, $\mathbf{v}_2 = (2,0,0)$, $\mathbf{v}_3 = (0,1,0)$ the set is dependent but $\mathbf{v}_3 \neq k_1\mathbf{v}_1 + k_2\mathbf{v}_2$.

Problem 2. Multiple choice [4 marks]

Each question is worth 1 mark. Tick exactly one option.

(a) Let $V = \mathbb{R}^2$ with the usual addition $(x_1,x_2) + (y_1,y_2) = (x_1 + y_1,\ x_2 + y_2)$ and the scalar multiplication $k(x_1,x_2) = (kx_1,\ 0)$. With these operations V is *not* a vector space: exactly one of the following four axioms fails. Which one?

□ A. Axiom 7: $k(\mathbf{u}+\mathbf{v}) = k\mathbf{u} + k\mathbf{v}$ ■ B. Axiom 10: $1\mathbf{u} = \mathbf{u}$

□ C. Axiom 8: $(k+m)\mathbf{u} = k\mathbf{u} + m\mathbf{u}$ □ D. Axiom 9: $k(m\mathbf{u}) = (km)\mathbf{u}$

Answer: B (Axiom 10). Take $\mathbf{u} = (7,8)$: $1\mathbf{u} = (1 \cdot 7,\ 0) = (7,0) \neq (7,8) = \mathbf{u}$, so Axiom 10 fails. The usual addition makes Axioms 7, 8, 9 hold: e.g. $(k+m)\mathbf{u} = ((k+m)u_1, 0) = (ku_1, 0) + (mu_1, 0) = k\mathbf{u} + m\mathbf{u}$.

(b) Which of the following subsets W is a **subspace** of $V = \mathbb{R}^2$ (with the usual operations)?

□ A. $W = \{(x,y) : x + y = 1\}$ □ B. $W = \{(x,y) : y = x^2\}$

■ C. $W = \{(x,y) : y = 3x\}$ □ D. $W = \{(x,y) : x \geq 0,\ y \geq 0\}$

Answer: C. A: $\mathbf{0} \notin W$. B: $(1,1) \in W$ but $2 \cdot (1,1) \notin W$. C: subspace (passes the Subspace Test). D: $(5,7) \in W$ but $-1 \cdot (5,7) \notin W$.

(c) Let $\mathbf{v}_1 = (1,0,2)$ and $\mathbf{v}_2 = (0,1,-1)$ in $\mathbb{R}^3$. Which of the following vectors belongs to $\text{span}\{\mathbf{v}_1, \mathbf{v}_2\}$?

□ A. $(3,1,7)$ ■ B. $(3,1,5)$ □ C. $(1,1,2)$ □ D. $(2,2,0)$

Answer: B. Match $k_1\mathbf{v}_1 + k_2\mathbf{v}_2$ with each candidate (two components fix k_1, k_2; the third must agree). A: $k_1 = 3$, $k_2 = 1$ would force the third component to be $5 \neq 7$. **B**: $k_1 = 3$, $k_2 = 1$ give third component $= 5$ ✓, so $(3,1,5) = 3\mathbf{v}_1 + \mathbf{v}_2$. C: $k_1 = 1$, $k_2 = 1$ would force the third component to be $1 \neq 2$. D: $k_1 = 2$, $k_2 = 2$ would force the third component to be $2 \neq 0$.

(d) Let $f_1 = x$ and $f_2 = e^x$ be functions in $C^\infty(-\infty, \infty)$. Compute the Wronskian $W(x)$ of f_1, f_2 and choose the correct statement.

□ A. $W(x) = (x-1)e^x$; the set is linearly dependent because $W(1) = 0$ ■ B. $W(x) = (x-1)e^x$; the set is linearly independent

□ C. $W(x) = (x+1)e^x$; the set is linearly independent □ D. $W(x) = 0$ for all x; the Wronskian test is inconclusive

Answer: B. $W(x) = \begin{vmatrix} x & e^x \\ 1 & e^x \end{vmatrix} = xe^x - e^x = (x-1)e^x$. This is not identically zero (e.g. $W(0) = -1$), so by Fact 4.4.2 the set is linearly independent.

Problem 3. Linear independence and span [4 marks]

Let $S = \{\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3\}$ be the set of vectors in $\mathbb{R}^3$ given by

$$\mathbf{v}_1 = (1, 2, -1), \qquad \mathbf{v}_2 = (2, 1, 1), \qquad \mathbf{v}_3 = (5, 4, 1),$$

and let $\mathbf{u} = (1, 1, 3)$.

(a) [2.5 marks] Determine whether S is linearly independent or linearly dependent. Show all your work.

(b) [1.5 marks] Determine whether $\mathbf{u}$ belongs to $\text{span}(S)$, that is, whether $\mathbf{u}$ is a linear combination of $\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3$. If it is, write $\mathbf{u}$ as such a linear combination; if it is not, explain why. (You may reuse the row operations from part (a).)

Solution.

(a) Set $k_1\mathbf{v}_1 + k_2\mathbf{v}_2 + k_3\mathbf{v}_3 = \mathbf{0}$, i.e. $k_1(1,2,-1) + k_2(2,1,1) + k_3(5,4,1) = (0,0,0)$. Comparing components gives the homogeneous system and its augmented matrix:

$$\begin{cases} k_1 + 2k_2 + 5k_3 = 0 \\ 2k_1 + k_2 + 4k_3 = 0 \\ -k_1 + k_2 + k_3 = 0 \end{cases} \qquad \left[\begin{array}{ccc|c} 1 & 2 & 5 & 0 \\ 2 & 1 & 4 & 0 \\ -1 & 1 & 1 & 0 \end{array}\right] \xrightarrow[R_1+R_3\to R_3]{-2R_1+R_2\to R_2} \left[\begin{array}{ccc|c} 1 & 2 & 5 & 0 \\ 0 & -3 & -6 & 0 \\ 0 & 3 & 6 & 0 \end{array}\right] \xrightarrow{R_2+R_3\to R_3} \left[\begin{array}{ccc|c} 1 & 2 & 5 & 0 \\ 0 & -3 & -6 & 0 \\ 0 & 0 & 0 & 0 \end{array}\right]$$

The coefficient matrix has only two pivots: the third row is a zero row, so k_3 is a free variable and the homogeneous system has **nontrivial solutions**. For instance $k_3 = 1$ gives $k_2 = -2$ and $k_1 = -1$, so $-\mathbf{v}_1 - 2\mathbf{v}_2 + \mathbf{v}_3 = \mathbf{0}$, i.e. $\mathbf{v}_3 = \mathbf{v}_1 + 2\mathbf{v}_2$.

S is linearly dependent.

(b) Set $k_1\mathbf{v}_1 + k_2\mathbf{v}_2 + k_3\mathbf{v}_3 = \mathbf{u}$ with $\mathbf{u} = (1,1,3)$. The coefficient matrix is the same, so the same row operations apply to the new right-hand side:

$$\left[\begin{array}{ccc|c} 1 & 2 & 5 & 1 \\ 2 & 1 & 4 & 1 \\ -1 & 1 & 1 & 3 \end{array}\right] \xrightarrow[R_1+R_3\to R_3]{-2R_1+R_2\to R_2} \left[\begin{array}{ccc|c} 1 & 2 & 5 & 1 \\ 0 & -3 & -6 & -1 \\ 0 & 3 & 6 & 4 \end{array}\right] \xrightarrow{R_2+R_3\to R_3} \left[\begin{array}{ccc|c} 1 & 2 & 5 & 1 \\ 0 & -3 & -6 & -1 \\ 0 & 0 & 0 & 3 \end{array}\right]$$

The last row reads $0k_1 + 0k_2 + 0k_3 = 3$, i.e. $0 = 3$, which is impossible: the system is **inconsistent**. Hence $\mathbf{u}$ is not a linear combination of $\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3$.

$\mathbf{u} \notin \text{span}(S)$.

Source: Example 4.4.1 (Linear Dependence in $\mathbb{R}^3$), line 13280; Example 4.2.10 (Vector IS a Linear Combination), line 11774; Example 4.2.11 (Vector Is NOT a Linear Combination), line 11843; Definition 4.4.1 (Linear Independence and Dependence), line 13236; Definition 4.3.1 (Span), line 12614 (`math231_lecture_notes_v02_2026_fully_solved.tex`). New vectors/polynomials; method, structure and hypotheses unchanged.

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Sections 4.1–4.4 (Real Vector Spaces, Subspaces, Spanning Sets, Linear Independence)

Fall 2026 • Thursday, 5 November 2026 • Time: 20 minutes • Total: 10 marks

Name: _____ ID: _____ Section: _____ **Version B**

Tick exactly one box for each True/False statement and exactly one option for each multiple-choice question; an item with more than one answer receives zero. Show all your work in Problem 3. Programmable calculators and mobile phones are not allowed.

Problem 1. True or False [2 marks]

Each statement is worth 0.5 mark. Tick **True** if the statement is always true, otherwise tick **False**. No justification is needed.

	True	False
(a) A set with two operations that satisfies nine of the ten vector space axioms is still a vector space.	☐	■
False. All ten axioms must hold (Definition 4.1.1); if even one axiom fails the set is not a vector space (Examples 4.1.3 and 4.1.4 fail a single axiom).		
(b) If A is an $m \times n$ matrix, then the solution set of the homogeneous system $A\mathbf{x} = \mathbf{0}$ is a subspace of $\mathbb{R}^n$.	■	☐
True. Example 4.2.7: $A\mathbf{0} = \mathbf{0}$, $A(\mathbf{u}+\mathbf{v}) = \mathbf{0}$ and $A(k\mathbf{u}) = \mathbf{0}$.		
(c) $\text{span}\{1, x, x^2\} = P_3$.	☐	■
False. Every linear combination $k_1 + k_2x + k_3x^2$ has no x^3 term, so $x^3 \in P_3$ is not in the span; the span is P_2 (Remark 4.3.2).		
(d) Every set of four vectors in $\mathbb{R}^3$ is linearly dependent.	■	☐
True. Fact 4.4.1(1) with $r = 4 > n = 3$ (Example 4.4.3).		

Problem 2. Multiple choice [4 marks]

Each question is worth 1 mark. Tick exactly one option.

(a) Let $V = \mathbb{R}^2$ with the usual addition $(x_1, x_2) + (y_1, y_2) = (x_1 + y_1,\ x_2 + y_2)$ and the scalar multiplication $k(x_1, x_2) = (kx_1,\ x_2)$. With these operations V is *not* a vector space: exactly one of the following four axioms fails. Which one?

☐ A. Axiom 10: $1\mathbf{u} = \mathbf{u}$

☐ B. Axiom 7: $k(\mathbf{u}+\mathbf{v}) = k\mathbf{u} + k\mathbf{v}$

■ C. Axiom 8: $(k+m)\mathbf{u} = k\mathbf{u} + m\mathbf{u}$

☐ D. Axiom 6: $k\mathbf{u}$ is in V (closure under scalar multiplication)

Answer: C (Axiom 8). Take $\mathbf{u} = (2, 5)$, $k = 2$, $m = 3$: $(k+m)\mathbf{u} = 5(2, 5) = (10, 5)$ but $k\mathbf{u} + m\mathbf{u} = (4, 5) + (6, 5) = (10, 10)$, so Axiom 8 fails. Axiom 10: $1\mathbf{u} = (u_1, u_2) = \mathbf{u}$ holds; Axiom 7: $k(\mathbf{u}+\mathbf{v}) = (k(u_1+v_1), u_2+v_2) = k\mathbf{u}+k\mathbf{v}$ holds; Axiom 6 holds since $(ku_1, u_2) \in \mathbb{R}^2$.

(b) Which of the following subsets of $V = M_{22}$ (with the usual operations) is **not** a subspace of V?

☐ A. all symmetric matrices ($A^T = A$)

■ B. all invertible matrices

☐ C. all matrices with $\text{tr}(A) = 0$

☐ D. all matrices $\begin{bmatrix} a & b \\ c & d \end{bmatrix}$ with $a = 0$

Answer: B. A: subspace (passes the Subspace Test). B: $\mathbf{0} \notin W$. C: subspace (passes the Subspace Test). D: subspace (passes the Subspace Test).

(c) Let $p_1 = 1 + x^2$ and $p_2 = x$ in P_2. Which of the following polynomials belongs to $\text{span}\{p_1, p_2\}$?

☐ A. $3 + 2x - 3x^2$ ☐ B. $2 + 3x + 3x^2$ ■ C. $3 + 2x + 3x^2$ ☐ D. $3 + 3x + 2x^2$

Answer: C. Match $k_1p_1 + k_2p_2$ with each candidate (two coefficients fix k_1, k_2; the third must agree). A: $k_1 = 3$, $k_2 = 2$ would force the x^2 coefficient to be $3 \neq -3$. B: $k_1 = 2$, $k_2 = 3$ would force the x^2 coefficient to be $2 \neq 3$. **C**: $k_1 = 3$, $k_2 = 2$ give x^2 coefficient $= 3$ ✓, so $3 + 2x + 3x^2 = 3p_1 + 2p_2$. D: $k_1 = 3$, $k_2 = 3$ would force the x^2 coefficient to be $3 \neq 2$.

(d) Let $f_1 = e^x$ and $f_2 = e^{2x}$ be functions in $C^\infty(-\infty, \infty)$. Compute the Wronskian $W(x)$ of f_1, f_2 and choose the correct statement.

☐ A. $W(x) = 3e^{3x}$; the set is linearly independent

☐ B. $W(x) = 0$ for all x; the Wronskian test is inconclusive

☐ C. $W(x) = e^{3x}$; the set is linearly dependent because e^{2x} is a multiple of e^x

■ D. $W(x) = e^{3x}$; the set is linearly independent

Answer: D. $W(x) = \begin{vmatrix} e^x & e^{2x} \\ e^x & 2e^{2x} \end{vmatrix} = 2e^{3x} - e^{3x} = e^{3x} \neq 0$ for every x, so the set is linearly independent (Fact 4.4.2; compare Example 4.4.5).

Problem 3. Linear independence and span [4 marks]

Let $S = \{p_1, p_2, p_3\}$ be the set of polynomials in P_2 given by

$$p_1 = 1 + x + x^2, \qquad p_2 = 1 - x + 2x^2, \qquad p_3 = 2 + x - x^2,$$

and let $u = 3 - 2x + 2x^2$.

(a) [2.5 marks] Determine whether S is linearly independent or linearly dependent. Show all your work.

(b) [1.5 marks] Determine whether u belongs to span(S), that is, whether u is a linear combination of p_1, p_2, p_3. If it is, write u as such a linear combination; if it is not, explain why. (You may reuse the row operations from part (a).)

Solution.

(a) Set $k_1p_1 + k_2p_2 + k_3p_3 = 0$, i.e. $k_1(1 + x + x^2) + k_2(1 - x + 2x^2) + k_3(2 + x - x^2) = 0 + 0x + 0x^2$. Comparing coefficients of 1, x and x^2 gives the homogeneous system and its augmented matrix:

$$\begin{cases} k_1 + k_2 + 2k_3 = 0 \\ k_1 - k_2 + k_3 = 0 \\ k_1 + 2k_2 - k_3 = 0 \end{cases} \qquad \left[\begin{array}{ccc|c} 1 & 1 & 2 & 0 \\ 1 & -1 & 1 & 0 \\ 1 & 2 & -1 & 0 \end{array}\right] \xrightarrow[-R_1+R_3\to R_3]{-R_1+R_2\to R_2} \left[\begin{array}{ccc|c} 1 & 1 & 2 & 0 \\ 0 & -2 & -1 & 0 \\ 0 & 1 & -3 & 0 \end{array}\right] \xrightarrow{R_2\leftrightarrow R_3} \left[\begin{array}{ccc|c} 1 & 1 & 2 & 0 \\ 0 & 1 & -3 & 0 \\ 0 & -2 & -1 & 0 \end{array}\right]$$

$$\xrightarrow{2R_2+R_3\to R_3} \left[\begin{array}{ccc|c} 1 & 1 & 2 & 0 \\ 0 & 1 & -3 & 0 \\ 0 & 0 & -7 & 0 \end{array}\right]$$

There are three pivots and no free variables, so the homogeneous system has **only the trivial solution** $k_1 = k_2 = k_3 = 0$.

S is linearly independent.

(b) Set $k_1p_1 + k_2p_2 + k_3p_3 = u$ with $u = 3 - 2x + 2x^2$. The coefficient matrix is the same, so the same row operations apply to the new right-hand side:

$$\left[\begin{array}{ccc|c} 1 & 1 & 2 & 3 \\ 1 & -1 & 1 & -2 \\ 1 & 2 & -1 & 2 \end{array}\right] \xrightarrow[-R_1+R_3\to R_3]{-R_1+R_2\to R_2} \left[\begin{array}{ccc|c} 1 & 1 & 2 & 3 \\ 0 & -2 & -1 & -5 \\ 0 & 1 & -3 & -1 \end{array}\right] \xrightarrow{R_2\leftrightarrow R_3} \left[\begin{array}{ccc|c} 1 & 1 & 2 & 3 \\ 0 & 1 & -3 & -1 \\ 0 & -2 & -1 & -5 \end{array}\right]$$

$$\xrightarrow{2R_2+R_3\to R_3} \left[\begin{array}{ccc|c} 1 & 1 & 2 & 3 \\ 0 & 1 & -3 & -1 \\ 0 & 0 & -7 & -7 \end{array}\right]$$

Back-substitution: $-7k_3 = -7 \Rightarrow k_3 = 1$; $k_2 - 3k_3 = -1 \Rightarrow k_2 = 2$; $k_1 + k_2 + 2k_3 = 3 \Rightarrow k_1 = -1$. Hence $u = -p_1 + 2p_2 + p_3$.

$u \in \text{span}(S)$: $u = -p_1 + 2p_2 + p_3$.

Source: Example 4.4.2 (Linear Independence in P_3), line 13357; Example 4.2.10 (Vector IS a Linear Combination), line 11774; Example 4.2.11 (Vector Is NOT a Linear Combination), line 11843; Definition 4.4.1 (Linear Independence and Dependence), line 13236; Definition 4.3.1 (Span), line 12614 (`math231_lecture_notes_v02_2026_fully_solved.tex`). New vectors/polynomials; method, structure and hypotheses unchanged.

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Math 231 – Linear Algebra Quiz 3

Sections 4.1–4.4 (Real Vector Spaces, Subspaces, Spanning Sets, Linear Independence)

Fall 2026 • Thursday, 5 November 2026 • Time: 20 minutes • Total: 10 marks

Name: _____ ID: _____ Section: _____ **Version C**

Tick exactly one box for each True/False statement and exactly one option for each multiple-choice question; an item with more than one answer receives zero. Show all your work in Problem 3. Programmable calculators and mobile phones are not allowed.

Problem 1. True or False [2 marks]

Each statement is worth 0.5 mark. Tick **True** if the statement is always true, otherwise tick **False**. No justification is needed.

	True	False
(a) If $\mathbf{u}$ is a vector in a vector space and k is a scalar with $k\mathbf{u} = \mathbf{0}$, then $k = 0$ or $\mathbf{u} = \mathbf{0}$.	■	□
True. This is Theorem 4.1.1(d).		
(b) If W_1 and W_2 are subspaces of a vector space V, then $W_1 \cup W_2$ is always a subspace of V.	□	■
False. Remark 4.2.2 / Example 4.2.9: the union of the x-axis and the y-axis is not closed under addition, $(0,7) + (2,0) = (2,7)$.		
(c) The vectors $(1,0,0)$ and $(0,1,0)$ span $\mathbb{R}^3$.	□	■
False. $k_1(1,0,0) + k_2(0,1,0) = (k_1, k_2, 0)$ always has third component 0, so e.g. $(0,0,1)$ is not a linear combination (Definition 4.3.1).		
(d) A set with exactly two vectors is linearly independent if and only if neither vector is a scalar multiple of the other.	■	□
True. Theorem 4.4.1(c).		

Problem 2. Multiple choice [4 marks]

Each question is worth 1 mark. Tick exactly one option.

(a) Let $V = \mathbb{R}^2$ with the usual addition $(x_1, x_2) + (y_1, y_2) = (x_1 + y_1,\ x_2 + y_2)$ and the scalar multiplication $k(x_1, x_2) = (k^2 x_1,\ k^2 x_2)$. With these operations V is *not* a vector space: exactly one of the following four axioms fails. Which one?

■ A. Axiom 8: $(k+m)\mathbf{u} = k\mathbf{u} + m\mathbf{u}$ □ B. Axiom 9: $k(m\mathbf{u}) = (km)\mathbf{u}$

□ C. Axiom 10: $1\mathbf{u} = \mathbf{u}$ □ D. Axiom 7: $k(\mathbf{u} + \mathbf{v}) = k\mathbf{u} + k\mathbf{v}$

Answer: A (Axiom 8). Take $\mathbf{u} = (1,1)$, $k = m = 1$: $(k+m)\mathbf{u} = 2(1,1) = (4,4)$ but $k\mathbf{u} + m\mathbf{u} = (1,1) + (1,1) = (2,2)$, so Axiom 8 fails. Axiom 9: $k(m\mathbf{u}) = k^2 m^2 \mathbf{u} = (km)^2\mathbf{u} = (km)\mathbf{u}$ holds; Axiom 10: $1^2\mathbf{u} = \mathbf{u}$ holds; Axiom 7: $k^2(\mathbf{u} + \mathbf{v}) = k^2\mathbf{u} + k^2\mathbf{v}$ holds.

(b) Which of the following subsets W is a **subspace** of $V = P_2$ (with the usual operations)?

□ A. $W = \{p \in P_2 : \deg(p) = 2\}$ □ B. $W = \{a + bx + cx^2 : a, b, c \text{ are integers}\}$

□ C. $W = \{p \in P_2 : p(0) = 1\}$ ■ D. $W = \{a + bx + cx^2 : a + b + c = 0\}$

Answer: D. A: $\mathbf{0} \notin W$. B: $1 + x + x^2 \in W$ but $\frac{1}{2} \cdot 1 + x + x^2 \notin W$. C: $\mathbf{0} \notin W$. D: subspace (passes the Subspace Test).

(c) Let $\mathbf{v}_1 = (1, 0, -1)$ and $\mathbf{v}_2 = (0, 1, 3)$ in $\mathbb{R}^3$. Which of the following vectors belongs to $\text{span}\{\mathbf{v}_1, \mathbf{v}_2\}$?

■ A. $(2, 1, 1)$ □ B. $(2, 1, 5)$ □ C. $(1, 2, -5)$ □ D. $(2, 1, -1)$

Answer: A. Match $k_1\mathbf{v}_1 + k_2\mathbf{v}_2$ with each candidate (two components fix k_1, k_2; the third must agree). **A**: $k_1 = 2$, $k_2 = 1$ give third component $= 1$ ✓, so $(2,1,1) = 2\mathbf{v}_1 + \mathbf{v}_2$. B: $k_1 = 2$, $k_2 = 1$ would force the third component to be $1 \neq 5$. C: $k_1 = 1$, $k_2 = 2$ would force the third component to be $5 \neq -5$. D: $k_1 = 2$, $k_2 = 1$ would force the third component to be $1 \neq -1$.

(d) Let $f_1 = 1$ and $f_2 = e^{-x}$ be functions in $C^\infty(-\infty, \infty)$. Compute the Wronskian $W(x)$ of f_1, f_2 and choose the correct statement.

■ A. $W(x) = -e^{-x}$; the set is linearly independent □ B. $W(x) = e^{-x}$; the set is linearly independent

□ C. $W(x) = -e^{-x}$; the set is linearly dependent because $W(x) < 0$ □ D. $W(x) = 0$ for all x; the Wronskian test is inconclusive

Answer: A. $W(x) = \begin{vmatrix} 1 & e^{-x} \\ 0 & -e^{-x} \end{vmatrix} = -e^{-x} \neq 0$ for every x, hence linearly independent (Fact 4.4.2).

Problem 3. Linear independence and span [4 marks]

Let $S = \{\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3\}$ be the set of vectors in $\mathbb{R}^3$ given by

$$\mathbf{v}_1 = (1, 1, 2), \qquad \mathbf{v}_2 = (1, 0, 1), \qquad \mathbf{v}_3 = (2, 1, 1),$$

and let $\mathbf{u} = (3, 3, 4)$.

(a) [2.5 marks] Determine whether S is linearly independent or linearly dependent. Show all your work.

(b) [1.5 marks] Determine whether $\mathbf{u}$ belongs to $\text{span}(S)$, that is, whether $\mathbf{u}$ is a linear combination of $\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3$. If it is, write $\mathbf{u}$ as such a linear combination; if it is not, explain why. (You may reuse the row operations from part (a).)

Solution.

(a) Set $k_1\mathbf{v}_1 + k_2\mathbf{v}_2 + k_3\mathbf{v}_3 = \mathbf{0}$, i.e. $k_1(1,1,2) + k_2(1,0,1) + k_3(2,1,1) = (0,0,0)$. Comparing components gives the homogeneous system and its augmented matrix:

$$\begin{cases} k_1 + k_2 + 2k_3 = 0 \\ k_1 + k_3 = 0 \\ 2k_1 + k_2 + k_3 = 0 \end{cases} \qquad \left[\begin{array}{ccc|c} 1 & 1 & 2 & 0 \\ 1 & 0 & 1 & 0 \\ 2 & 1 & 1 & 0 \end{array}\right] \xrightarrow[-2R_1+R_3\to R_3]{-R_1+R_2\to R_2} \left[\begin{array}{ccc|c} 1 & 1 & 2 & 0 \\ 0 & -1 & -1 & 0 \\ 0 & -1 & -3 & 0 \end{array}\right] \xrightarrow{-R_2+R_3\to R_3} \left[\begin{array}{ccc|c} 1 & 1 & 2 & 0 \\ 0 & -1 & -1 & 0 \\ 0 & 0 & -2 & 0 \end{array}\right]$$

There are three pivots and no free variables, so the homogeneous system has **only the trivial solution** $k_1 = k_2 = k_3 = 0$.

S is linearly independent.

(b) Set $k_1\mathbf{v}_1 + k_2\mathbf{v}_2 + k_3\mathbf{v}_3 = \mathbf{u}$ with $\mathbf{u} = (3,3,4)$. The coefficient matrix is the same, so the same row operations apply to the new right-hand side:

$$\left[\begin{array}{ccc|c} 1 & 1 & 2 & 3 \\ 1 & 0 & 1 & 3 \\ 2 & 1 & 1 & 4 \end{array}\right] \xrightarrow[-2R_1+R_3\to R_3]{-R_1+R_2\to R_2} \left[\begin{array}{ccc|c} 1 & 1 & 2 & 3 \\ 0 & -1 & -1 & 0 \\ 0 & -1 & -3 & -2 \end{array}\right] \xrightarrow{-R_2+R_3\to R_3} \left[\begin{array}{ccc|c} 1 & 1 & 2 & 3 \\ 0 & -1 & -1 & 0 \\ 0 & 0 & -2 & -2 \end{array}\right]$$

Back-substitution: $-2k_3 = -2 \Rightarrow k_3 = 1$; $-k_2 - k_3 = 0 \Rightarrow k_2 = -1$; $k_1 + k_2 + 2k_3 = 3 \Rightarrow k_1 = 2$. Hence $\mathbf{u} = 2\mathbf{v}_1 - \mathbf{v}_2 + \mathbf{v}_3$.

$\mathbf{u} \in \text{span}(S)$: $\mathbf{u} = 2\mathbf{v}_1 - \mathbf{v}_2 + \mathbf{v}_3$.

Source: Example 4.4.1 (Linear Dependence in $\mathbb{R}^3$), line 13280; Example 4.2.10 (Vector IS a Linear Combination), line 11774; Example 4.2.11 (Vector Is NOT a Linear Combination), line 11843; Definition 4.4.1 (Linear Independence and Dependence), line 13236; Definition 4.3.1 (Span), line 12614 (`math231_lecture_notes_v02_2026_fully_solved.tex`). New vectors/polynomials; method, structure and hypotheses unchanged.

Qatar University

Department of Mathematics, Statistics and Physics

Department of Mathematics and Statistics

Math 231 – Linear Algebra Quiz 3

Sections 4.1–4.4 (Real Vector Spaces, Subspaces, Spanning Sets, Linear Independence)

Fall 2026 • Thursday, 5 November 2026 • Time: 20 minutes • Total: 10 marks

Name: _____ ID: _____ Section: _____ **Version D**

Tick exactly one box for each True/False statement and exactly one option for each multiple-choice question; an item with more than one answer receives zero. Show all your work in Problem 3. Programmable calculators and mobile phones are not allowed.

Problem 1. True or False [2 marks]

Each statement is worth 0.5 mark. Tick **True** if the statement is always true, otherwise tick **False**. No justification is needed.

	True	False
(a) In a vector space, if $k\mathbf{u} = \mathbf{0}$ for a scalar k and a vector $\mathbf{u}$, then both $k = 0$ and $\mathbf{u} = \mathbf{0}$.	☐	■
False. Theorem 4.1.1(d) gives $k = 0$ *or* $\mathbf{u} = \mathbf{0}$, not both: e.g. $0\mathbf{u} = \mathbf{0}$ for $\mathbf{u} = (1, 1) \neq \mathbf{0}$.		
(b) If W_1 and W_2 are subspaces of a vector space V, then $W_1 \cap W_2$ is a subspace of V.	■	☐
True. Theorem 4.2.2 (intersection of subspaces).		
(c) Every vector in a set S belongs to span(S).	■	☐
True. Remark 4.3.1(1): $\mathbf{w}_i = 0\mathbf{w}_1 + \cdots + 1\mathbf{w}_i + \cdots + 0\mathbf{w}_r$.		
(d) The functions e^x and $6e^x$ form a linearly independent set in $C(-\infty, \infty)$.	☐	■
False. $6e^x$ is a scalar multiple of e^x, so the set is linearly dependent (Example 4.4.7, Theorem 4.4.1(c)).		

Problem 2. Multiple choice [4 marks]

Each question is worth 1 mark. Tick exactly one option.

(a) Let $V = \mathbb{R}^2$ with the usual addition $(x_1, x_2) + (y_1, y_2) = (x_1 + y_1,\ x_2 + y_2)$ and the scalar multiplication $k(x_1, x_2) = (0,\ kx_2)$. With these operations V is *not* a vector space: exactly one of the following four axioms fails. Which one?

☐ A. Axiom 6: $k\mathbf{u}$ is in V (closure under scalar multiplication)

☐ B. Axiom 4: there is a zero vector $\mathbf{0}$ with $\mathbf{0} + \mathbf{u} = \mathbf{u}$

☐ C. Axiom 7: $k(\mathbf{u} + \mathbf{v}) = k\mathbf{u} + k\mathbf{v}$

■ D. Axiom 10: $1\mathbf{u} = \mathbf{u}$

Answer: D (Axiom 10). Take $\mathbf{u} = (3, 4)$: $1\mathbf{u} = (0, 4) \neq (3, 4) = \mathbf{u}$, so Axiom 10 fails. Axiom 6 holds since $(0, ku_2) \in \mathbb{R}^2$; Axiom 4 holds with $\mathbf{0} = (0, 0)$ (usual addition); Axiom 7: $k(\mathbf{u} + \mathbf{v}) = (0, k(u_2 + v_2)) = (0, ku_2) + (0, kv_2)$ holds.

(b) Which of the following subsets of $V = \mathbb{R}^3$ (with the usual operations) is **not** a subspace of V?

■ A. $W = \{(x, y, z) : x + y + z = 1\}$

☐ B. $W = \{(x, y, z) : x + y + z = 0\}$

☐ C. $W = \{(x, y, z) : x = y = z\}$

☐ D. $W = \{(x, y, z) : x = 2y \text{ and } z = 0\}$

Answer: A. A: $\mathbf{0} \notin W$. B: subspace (passes the Subspace Test). C: subspace (passes the Subspace Test). D: subspace (passes the Subspace Test).

(c) Let $p_1 = 1 + x$ and $p_2 = x^2$ in P_2. Which of the following polynomials belongs to span$\{p_1, p_2\}$?

☐ A. $2 - 2x - x^2$ ☐ B. $2 + x - x^2$ ☐ C. $1 + 2x + 2x^2$ ■ D. $2 + 2x - x^2$

Answer: D. Match $k_1 p_1 + k_2 p_2$ with each candidate (two coefficients fix k_1, k_2; the third must agree). A: $k_1 = 2$, $k_2 = -1$ would force the x coefficient to be $2 \neq -2$. B: $k_1 = 2$, $k_2 = -1$ would force the x coefficient to be $2 \neq 1$. C: $k_1 = 1$, $k_2 = 2$ would force the x coefficient to be $1 \neq 2$. **D**: $k_1 = 2$, $k_2 = -1$ give x coefficient $= 2$ ✓, so $2 + 2x - x^2 = 2p_1 - p_2$.

(d) Let $f_1 = x$ and $f_2 = x^2$ be functions in $C^\infty(-\infty, \infty)$. Compute the Wronskian $W(x)$ of f_1, f_2 and choose the correct statement.

☐ A. $W(x) = x^2$; the set is linearly dependent because $W(0) = 0$

☐ B. $W(x) = x^2$; the Wronskian test is inconclusive because $W(0) = 0$

■ C. $W(x) = x^2$; the set is linearly independent

☐ D. $W(x) = 3x^2$; the set is linearly independent

Answer: C. $W(x) = \begin{vmatrix} x & x^2 \\ 1 & 2x \end{vmatrix} = 2x^2 - x^2 = x^2$. Since $W(1) = 1 \neq 0$, W is not identically zero and the set is linearly independent (Fact 4.4.2).

Problem 3. Linear independence and span [4 marks]

Let $S = \{p_1, p_2, p_3\}$ be the set of polynomials in P_2 given by

$$p_1 = 1 + x, \qquad p_2 = 1 - x + x^2, \qquad p_3 = 1 + 3x - x^2,$$

and let $u = 1 + 2x + x^2$.

(a) [2.5 marks] Determine whether S is linearly independent or linearly dependent. Show all your work.

(b) [1.5 marks] Determine whether u belongs to $\text{span}(S)$, that is, whether u is a linear combination of p_1, p_2, p_3. If it is, write u as such a linear combination; if it is not, explain why. (You may reuse the row operations from part (a).)

Solution.

(a) Set $k_1p_1 + k_2p_2 + k_3p_3 = 0$, i.e. $k_1(1+x) + k_2(1-x+x^2) + k_3(1+3x-x^2) = 0 + 0x + 0x^2$. Comparing coefficients of 1, x and x^2 gives the homogeneous system and its augmented matrix:

$$\begin{cases} k_1 + k_2 + k_3 = 0 \\ k_1 - k_2 + 3k_3 = 0 \\ k_2 - k_3 = 0 \end{cases} \qquad \left[\begin{array}{ccc|c} 1 & 1 & 1 & 0 \\ 1 & -1 & 3 & 0 \\ 0 & 1 & -1 & 0 \end{array}\right] \xrightarrow{-R_1+R_2\to R_2} \left[\begin{array}{ccc|c} 1 & 1 & 1 & 0 \\ 0 & -2 & 2 & 0 \\ 0 & 1 & -1 & 0 \end{array}\right] \xrightarrow{R_2\leftrightarrow R_3} \left[\begin{array}{ccc|c} 1 & 1 & 1 & 0 \\ 0 & 1 & -1 & 0 \\ 0 & -2 & 2 & 0 \end{array}\right]$$

$$\xrightarrow{2R_2+R_3\to R_3} \left[\begin{array}{ccc|c} 1 & 1 & 1 & 0 \\ 0 & 1 & -1 & 0 \\ 0 & 0 & 0 & 0 \end{array}\right]$$

The coefficient matrix has only two pivots: the third row is a zero row, so k_3 is a free variable and the homogeneous system has **nontrivial solutions**. For instance $k_3 = 1$ gives $k_2 = 1$ and $k_1 = -2$, so $-2p_1 + p_2 + p_3 = 0$, i.e. $p_3 = 2p_1 - p_2$.

S is linearly dependent.

(b) Set $k_1p_1 + k_2p_2 + k_3p_3 = u$ with $u = 1 + 2x + x^2$. The coefficient matrix is the same, so the same row operations apply to the new right-hand side:

$$\left[\begin{array}{ccc|c} 1 & 1 & 1 & 1 \\ 1 & -1 & 3 & 2 \\ 0 & 1 & -1 & 1 \end{array}\right] \xrightarrow{-R_1+R_2\to R_2} \left[\begin{array}{ccc|c} 1 & 1 & 1 & 1 \\ 0 & -2 & 2 & 1 \\ 0 & 1 & -1 & 1 \end{array}\right] \xrightarrow{R_2\leftrightarrow R_3} \left[\begin{array}{ccc|c} 1 & 1 & 1 & 1 \\ 0 & 1 & -1 & 1 \\ 0 & -2 & 2 & 1 \end{array}\right]$$

$$\xrightarrow{2R_2+R_3\to R_3} \left[\begin{array}{ccc|c} 1 & 1 & 1 & 1 \\ 0 & 1 & -1 & 1 \\ 0 & 0 & 0 & 3 \end{array}\right]$$

The last row reads $0k_1 + 0k_2 + 0k_3 = 3$, i.e. $0 = 3$, which is impossible: the system is **inconsistent**. Hence u is not a linear combination of p_1, p_2, p_3.

$u \notin \text{span}(S)$.

Source: Example 4.4.2 (Linear Independence in P_3), line 13357; Example 4.2.10 (Vector IS a Linear Combination), line 11774; Example 4.2.11 (Vector Is NOT a Linear Combination), line 11843; Definition 4.4.1 (Linear Independence and Dependence), line 13236; Definition 4.3.1 (Span), line 12614 (`math231_lecture_notes_v02_2026_fully_solved.tex`). New vectors/polynomials; method, structure and hypotheses unchanged.

Qatar University

Department of Mathematics, Statistics and Physics

Department of Mathematics and Statistics

Math 231 – Linear Algebra Quiz 3

Sections 4.1–4.4 (Real Vector Spaces, Subspaces, Spanning Sets, Linear Independence)

Fall 2026 • Thursday, 5 November 2026 • Time: 20 minutes • Total: 10 marks

Name: _____ ID: _____ Section: _____ **Version E**

Tick exactly one box for each True/False statement and exactly one option for each multiple-choice question; an item with more than one answer receives zero. Show all your work in Problem 3. Programmable calculators and mobile phones are not allowed.

Problem 1. True or False [2 marks]

Each statement is worth 0.5 mark. Tick **True** if the statement is always true, otherwise tick **False**. No justification is needed.

	True	False
(a) The set $F(-\infty,\infty)$ of all real-valued functions defined on $(-\infty,\infty)$, with pointwise addition and scalar multiplication, is a vector space.	■	□
True. Example 4.1.2 verifies all ten axioms for $F(-\infty,\infty)$.		
(b) The set of all $n\times n$ matrices A with $\operatorname{tr}(A)=0$ is a subspace of M_{nn}.	■	□
True. $\operatorname{tr}(O)=0$, $\operatorname{tr}(A+B)=\operatorname{tr}A+\operatorname{tr}B$, $\operatorname{tr}(kA)=k\operatorname{tr}A$ (Extra Problems 4.2.6).		
(c) If $\operatorname{span}(S)=\mathbb{R}^3$, then S must contain exactly three vectors.	□	■
False. Extra Problems 4.3.2: the four vectors $(1,-1,0),(1,2,1),(1,0,1),(3,3,-1)$ span $\mathbb{R}^3$.		
(d) If the Wronskian $W(x)$ of two functions satisfies $W(0)=0$, then the Wronskian test is inconclusive.	□	■
False. Only $W(x)\equiv 0$ (zero for *every* x) makes the test inconclusive; e.g. $W(x)=x^2$ has $W(0)=0$ but $W(1)=1\neq 0$, so the functions are linearly independent (Remark, item 2).		

Problem 2. Multiple choice [4 marks]

Each question is worth 1 mark. Tick exactly one option.

(a) Let $V=\mathbb{R}^3$ with the usual addition $(x_1,x_2,x_3)+(y_1,y_2,y_3)=(x_1+y_1,\ x_2+y_2,\ x_3+y_3)$ and the scalar multiplication $k(x_1,x_2,x_3)=(kx_1,\ x_2,\ x_3)$. With these operations V is *not* a vector space: exactly one of the following four axioms fails. Which one?

□ A. Axiom 9: $k(m\mathbf{u})=(km)\mathbf{u}$ ■ B. Axiom 8: $(k+m)\mathbf{u}=k\mathbf{u}+m\mathbf{u}$

□ C. Axiom 10: $1\mathbf{u}=\mathbf{u}$ □ D. Axiom 2: $\mathbf{u}+\mathbf{v}=\mathbf{v}+\mathbf{u}$

Answer: B (Axiom 8). Take $\mathbf{u}=(1,1,1)$, $k=m=1$: $(k+m)\mathbf{u}=2\mathbf{u}=(2,1,1)$ but $k\mathbf{u}+m\mathbf{u}=(1,1,1)+(1,1,1)=(2,2,2)$, so Axiom 8 fails. Axiom 9: $k(m\mathbf{u})=(kmu_1,u_2,u_3)=(km)\mathbf{u}$ holds; Axiom 10: $1\mathbf{u}=\mathbf{u}$ holds; Axiom 2 holds for the usual addition.

(b) Which of the following subsets W is a **subspace** of $V=F(-\infty,\infty)$ (with the usual operations)?

□ A. $W=\{f : f(0)=1\}$ □ B. $W=\{f : f(x)\geq 0 \text{ for all } x\}$

■ C. $W=C(-\infty,\infty)$, all continuous functions □ D. $W=\{f : f(1)=2\}$

Answer: C. A: $\mathbf{0}\notin W$. B: $f=1\in W$ but $(-1)f\notin W$. C: subspace (passes the Subspace Test). D: $\mathbf{0}\notin W$.

(c) Let $\mathbf{v}_1=(1,1,0)$ and $\mathbf{v}_2=(0,1,1)$ in $\mathbb{R}^3$. Which of the following vectors belongs to $\operatorname{span}\{\mathbf{v}_1,\mathbf{v}_2\}$?

□ A. $(2,3,5)$ ■ B. $(2,5,3)$ □ C. $(2,5,4)$ □ D. $(1,1,1)$

Answer: B. Match $k_1\mathbf{v}_1+k_2\mathbf{v}_2$ with each candidate (two components fix k_1,k_2; the third must agree). A: $k_1=2$, $k_2=1$ would force the third component to be $1\neq 5$. **B**: $k_1=2$, $k_2=3$ give third component $=3$ ✓, so $(2,5,3)=2\mathbf{v}_1+3\mathbf{v}_2$. C: $k_1=2$, $k_2=3$ would force the third component to be $3\neq 4$. D: $k_1=1$, $k_2=0$ would force the third component to be $0\neq 1$.

(d) Let $f_1=e^x$ and $f_2=xe^x$ be functions in $C^\infty(-\infty,\infty)$. Compute the Wronskian $W(x)$ of f_1,f_2 and choose the correct statement.

■ A. $W(x)=e^{2x}$; the set is linearly independent □ B. $W(x)=xe^{2x}$; the set is linearly independent

□ C. $W(x)=xe^{2x}$; the set is linearly dependent because $W(0)=0$ □ D. $W(x)=e^{2x}$; the set is linearly dependent because xe^x is a multiple of e^x

Answer: A. $(xe^x)'=e^x+xe^x$, so $W(x)=\begin{vmatrix} e^x & xe^x \\ e^x & e^x+xe^x \end{vmatrix}=e^{2x}+xe^{2x}-xe^{2x}=e^{2x}\neq 0$; linearly independent by Fact 4.4.2.

Problem 3. Linear independence and span [4 marks]

Let $S = \{\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3\}$ be the set of vectors in $\mathbb{R}^3$ given by

$$\mathbf{v}_1 = (1, -1, 2), \qquad \mathbf{v}_2 = (1, 1, 0), \qquad \mathbf{v}_3 = (1, -3, 4),$$

and let $\mathbf{u} = (1, 2, 1)$.

(a) [2.5 marks] Determine whether S is linearly independent or linearly dependent. Show all your work.

(b) [1.5 marks] Determine whether $\mathbf{u}$ belongs to $\text{span}(S)$, that is, whether $\mathbf{u}$ is a linear combination of $\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3$. If it is, write $\mathbf{u}$ as such a linear combination; if it is not, explain why. (You may reuse the row operations from part (a).)

Solution.

(a) Set $k_1\mathbf{v}_1 + k_2\mathbf{v}_2 + k_3\mathbf{v}_3 = \mathbf{0}$, i.e. $k_1(1,-1,2) + k_2(1,1,0) + k_3(1,-3,4) = (0,0,0)$. Comparing components gives the homogeneous system and its augmented matrix:

$$\begin{cases} k_1 + k_2 + k_3 = 0 \\ -k_1 + k_2 - 3k_3 = 0 \\ 2k_1 + 4k_3 = 0 \end{cases} \qquad \left[\begin{array}{ccc|c} 1 & 1 & 1 & 0 \\ -1 & 1 & -3 & 0 \\ 2 & 0 & 4 & 0 \end{array}\right] \xrightarrow[-2R_1+R_3\to R_3]{R_1+R_2\to R_2} \left[\begin{array}{ccc|c} 1 & 1 & 1 & 0 \\ 0 & 2 & -2 & 0 \\ 0 & -2 & 2 & 0 \end{array}\right] \xrightarrow{R_2+R_3\to R_3} \left[\begin{array}{ccc|c} 1 & 1 & 1 & 0 \\ 0 & 2 & -2 & 0 \\ 0 & 0 & 0 & 0 \end{array}\right]$$

The coefficient matrix has only two pivots: the third row is a zero row, so k_3 is a free variable and the homogeneous system has **nontrivial solutions**. For instance $k_3 = 1$ gives $k_2 = 1$ and $k_1 = -2$, so $-2\mathbf{v}_1 + \mathbf{v}_2 + \mathbf{v}_3 = \mathbf{0}$, i.e. $\mathbf{v}_3 = 2\mathbf{v}_1 - \mathbf{v}_2$.

S is linearly dependent.

(b) Set $k_1\mathbf{v}_1 + k_2\mathbf{v}_2 + k_3\mathbf{v}_3 = \mathbf{u}$ with $\mathbf{u} = (1, 2, 1)$. The coefficient matrix is the same, so the same row operations apply to the new right-hand side:

$$\left[\begin{array}{ccc|c} 1 & 1 & 1 & 1 \\ -1 & 1 & -3 & 2 \\ 2 & 0 & 4 & 1 \end{array}\right] \xrightarrow[-2R_1+R_3\to R_3]{R_1+R_2\to R_2} \left[\begin{array}{ccc|c} 1 & 1 & 1 & 1 \\ 0 & 2 & -2 & 3 \\ 0 & -2 & 2 & -1 \end{array}\right] \xrightarrow{R_2+R_3\to R_3} \left[\begin{array}{ccc|c} 1 & 1 & 1 & 1 \\ 0 & 2 & -2 & 3 \\ 0 & 0 & 0 & 2 \end{array}\right]$$

The last row reads $0k_1 + 0k_2 + 0k_3 = 2$, i.e. $0 = 2$, which is impossible: the system is **inconsistent**. Hence $\mathbf{u}$ is not a linear combination of $\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3$.

$\mathbf{u} \notin \text{span}(S)$.

Source: Example 4.4.1 (Linear Dependence in $\mathbb{R}^3$), line 13280; Example 4.2.10 (Vector IS a Linear Combination), line 11774; Example 4.2.11 (Vector Is NOT a Linear Combination), line 11843; Definition 4.4.1 (Linear Independence and Dependence), line 13236; Definition 4.3.1 (Span), line 12614 (`math231_lecture_notes_v02_2026_fully_solved.tex`). New vectors/polynomials; method, structure and hypotheses unchanged.

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Math 231 – Linear Algebra Quiz 3

Sections 4.1–4.4 (Real Vector Spaces, Subspaces, Spanning Sets, Linear Independence)

Fall 2026 • Thursday, 5 November 2026 • Time: 20 minutes • Total: 10 marks

Name: _____ ID: _____ Section: _____ **Version F**

Tick exactly one box for each True/False statement and exactly one option for each multiple-choice question; an item with more than one answer receives zero. Show all your work in Problem 3. Programmable calculators and mobile phones are not allowed.

Problem 1. True or False [2 marks]

Each statement is worth 0.5 mark. Tick **True** if the statement is always true, otherwise tick **False**. No justification is needed.

	True	False
(a) To show that a set with two operations is a vector space, it is enough to verify Axiom 1 (closure under addition) and Axiom 6 (closure under scalar multiplication).	☐	■
False. Closure is only Step 3 of the procedure; Step 4 requires Axioms 2, 3, 4, 5, 7, 8, 9, 10 as well. Example 4.1.3 is closed under both operations yet fails Axiom 10.		
(b) The first quadrant $W = \{(x, y) \in \mathbb{R}^2 : x \geq 0,\ y \geq 0\}$ is a subspace of $\mathbb{R}^2$.	☐	■
False. Not closed under scalar multiplication: $(-1)(5, 7) = (-5, -7) \notin W$ (Example 4.2.2).		
(c) $\text{span}\{1, x, x^2\} = P_2$.	■	☐
True. Remark 4.3.2(2): every $a + bx + cx^2$ is a linear combination of $1, x, x^2$.		
(d) If the Wronskian of $f_1, \ldots, f_n$ is not identically zero on $(-\infty, \infty)$, then the functions are linearly independent.	■	☐
True. Fact 4.4.2 (Wronskian test).		

Problem 2. Multiple choice [4 marks]

Each question is worth 1 mark. Tick exactly one option.

(a) Let $V = \mathbb{R}^3$ with the usual addition $(x_1, x_2, x_3) + (y_1, y_2, y_3) = (x_1 + y_1,\ x_2 + y_2,\ x_3 + y_3)$ and the scalar multiplication $k(x_1, x_2, x_3) = (kx_1,\ kx_2,\ 0)$. With these operations V is *not* a vector space: exactly one of the following four axioms fails. Which one?

☐ A. Axiom 1: $\mathbf{u} + \mathbf{v}$ is in V (closure under addition) ☐ B. Axiom 8: $(k + m)\mathbf{u} = k\mathbf{u} + m\mathbf{u}$

☐ C. Axiom 9: $k(m\mathbf{u}) = (km)\mathbf{u}$ ■ D. Axiom 10: $1\mathbf{u} = \mathbf{u}$

Answer: D (Axiom 10). Take $\mathbf{u} = (1, 2, 3)$: $1\mathbf{u} = (1, 2, 0) \neq \mathbf{u}$, so Axiom 10 fails. Axiom 1 holds (usual addition); Axiom 8: $(k + m)\mathbf{u} = ((k + m)u_1, (k + m)u_2, 0) = k\mathbf{u} + m\mathbf{u}$ holds; Axiom 9: $k(m\mathbf{u}) = (kmu_1, kmu_2, 0) = (km)\mathbf{u}$ holds.

(b) Which of the following subsets W is a **subspace** of $V = M_{22}$ (with the usual operations)?

☐ A. all matrices with $\det(A) = 0$ ■ B. all matrices with $\text{tr}(A) = 0$

☐ C. all matrices of the form $\begin{bmatrix} 1 & a \\ b & c \end{bmatrix}$ ☐ D. all matrices whose four entries are all ≥ 0

Answer: B. A: $\begin{bmatrix} 1 & 2 \\ 1 & 2 \end{bmatrix}, \begin{bmatrix} 3 & -1 \\ 6 & -2 \end{bmatrix} \in W$ but their sum $\notin W$. B: subspace (passes the Subspace Test). C: $\mathbf{0} \notin W$. D: $\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \in W$ but $-1 \cdot \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \notin W$.

(c) Let $p_1 = 1 - x$ and $p_2 = 1 + x^2$ in P_2. Which of the following polynomials belongs to $\text{span}\{p_1, p_2\}$?

☐ A. $1 + 2x - 3x^2$ ☐ B. $3 + 2x + x^2$ ■ C. $1 + 2x + 3x^2$ ☐ D. $1 - 2x + 3x^2$

Answer: C. Match $k_1 p_1 + k_2 p_2$ with each candidate (two coefficients fix k_1, k_2; the third must agree). A: $k_1 = -2, k_2 = 3$ would force the x^2 coefficient to be $3 \neq -3$. B: $k_1 = -2, k_2 = 5$ would force the x^2 coefficient to be $5 \neq 1$. **C**: $k_1 = -2, k_2 = 3$ give x^2 coefficient $= 3$ ✓, so $1 + 2x + 3x^2 = -2p_1 + 3p_2$. D: $k_1 = 2, k_2 = -1$ would force the x^2 coefficient to be $-1 \neq 3$.

(d) Let $f_1 = 1$ and $f_2 = x^2$ be functions in $C^\infty(-\infty, \infty)$. Compute the Wronskian $W(x)$ of f_1, f_2 and choose the correct statement.

☐ A. $W(x) = 2x$; the set is linearly dependent because $W(0) = 0$ ☐ B. $W(x) = -2x$; the set is linearly independent

☐ C. $W(x) = 2x$; the Wronskian test is inconclusive because $W(0) = 0$ ■ D. $W(x) = 2x$; the set is linearly independent

Answer: D. $W(x) = \begin{vmatrix} 1 & x^2 \\ 0 & 2x \end{vmatrix} = 2x$. $W(1) = 2 \neq 0$, so $W \not\equiv 0$ and the set is linearly independent (Fact 4.4.2).

Problem 3. Linear independence and span [4 marks]

Let $S = \{p_1, p_2, p_3\}$ be the set of polynomials in P_2 given by

$$p_1 = 1 + x^2, \qquad p_2 = x + x^2, \qquad p_3 = 1 - x + x^2,$$

and let $u = -1 + 4x + x^2$.

(a) [2.5 marks] Determine whether S is linearly independent or linearly dependent. Show all your work.

(b) [1.5 marks] Determine whether u belongs to span(S), that is, whether u is a linear combination of p_1, p_2, p_3. If it is, write u as such a linear combination; if it is not, explain why. (You may reuse the row operations from part (a).)

Solution.

(a) Set $k_1p_1 + k_2p_2 + k_3p_3 = 0$, i.e. $k_1(1 + x^2) + k_2(x + x^2) + k_3(1 - x + x^2) = 0 + 0x + 0x^2$. Comparing coefficients of 1, x and x^2 gives the homogeneous system and its augmented matrix:

$$\begin{cases} k_1 + k_3 = 0 \\ k_2 - k_3 = 0 \\ k_1 + k_2 + k_3 = 0 \end{cases} \qquad \left[\begin{array}{ccc|c} 1 & 0 & 1 & 0 \\ 0 & 1 & -1 & 0 \\ 1 & 1 & 1 & 0 \end{array}\right] \xrightarrow{-R_1+R_3\to R_3} \left[\begin{array}{ccc|c} 1 & 0 & 1 & 0 \\ 0 & 1 & -1 & 0 \\ 0 & 1 & 0 & 0 \end{array}\right] \xrightarrow{-R_2+R_3\to R_3} \left[\begin{array}{ccc|c} 1 & 0 & 1 & 0 \\ 0 & 1 & -1 & 0 \\ 0 & 0 & 1 & 0 \end{array}\right]$$

There are three pivots and no free variables, so the homogeneous system has **only the trivial solution** $k_1 = k_2 = k_3 = 0$.

S is linearly independent.

(b) Set $k_1p_1 + k_2p_2 + k_3p_3 = u$ with $u = -1 + 4x + x^2$. The coefficient matrix is the same, so the same row operations apply to the new right-hand side:

$$\left[\begin{array}{ccc|c} 1 & 0 & 1 & -1 \\ 0 & 1 & -1 & 4 \\ 1 & 1 & 1 & 1 \end{array}\right] \xrightarrow{-R_1+R_3\to R_3} \left[\begin{array}{ccc|c} 1 & 0 & 1 & -1 \\ 0 & 1 & -1 & 4 \\ 0 & 1 & 0 & 2 \end{array}\right] \xrightarrow{-R_2+R_3\to R_3} \left[\begin{array}{ccc|c} 1 & 0 & 1 & -1 \\ 0 & 1 & -1 & 4 \\ 0 & 0 & 1 & -2 \end{array}\right]$$

Back-substitution: $k_3 = -2 \Rightarrow k_3 = -2$; $k_2 - k_3 = 4 \Rightarrow k_2 = 2$; $k_1 + k_3 = -1 \Rightarrow k_1 = 1$. Hence $u = p_1 + 2p_2 - 2p_3$.

$u \in$ span(S): $u = p_1 + 2p_2 - 2p_3$.

Source: Example 4.4.2 (Linear Independence in P_3), line 13357; Example 4.2.10 (Vector IS a Linear Combination), line 11774; Example 4.2.11 (Vector Is NOT a Linear Combination), line 11843; Definition 4.4.1 (Linear Independence and Dependence), line 13236; Definition 4.3.1 (Span), line 12614 (`math231_lecture_notes_v02_2026_fully_solved.tex`). New vectors/polynomials; method, structure and hypotheses unchanged.

Qatar University

Department of Mathematics, Statistics and Physics

Department of Mathematics and Statistics

Math 231 – Linear Algebra Quiz 3

Sections 4.1–4.4 (Real Vector Spaces, Subspaces, Spanning Sets, Linear Independence)

Fall 2026 • Thursday, 5 November 2026 • Time: 20 minutes • Total: 10 marks

Name: _____ ID: _____ Section: _____ **Version G**

Tick exactly one box for each True/False statement and exactly one option for each multiple-choice question; an item with more than one answer receives zero. Show all your work in Problem 3. Programmable calculators and mobile phones are not allowed.

Problem 1. True or False [2 marks]

Each statement is worth 0.5 mark. Tick **True** if the statement is always true, otherwise tick **False**. No justification is needed.

	True	False
(a) In a vector space V, $(-1)\mathbf{u} = -\mathbf{u}$ for every vector $\mathbf{u}$ in V.	■	□
(b) The set $W = \{a + bx + cx^2 : a, b, c \text{ are integers}\}$ is a subspace of P_2.	□	■
(c) The zero vector belongs to $\text{span}(S)$ for every nonempty set S of vectors in V.	■	□
(d) If the Wronskian of $f_1, \ldots, f_n$ is identically zero on $(-\infty, \infty)$, then the functions are linearly dependent.	□	■

True. This is Theorem 4.1.1(c).

False. Not closed under scalar multiplication: $\frac{1}{2}(1 + x + x^2) \notin W$ (Extra Problems 4.2.4).

True. $\mathbf{0} = 0\mathbf{w}_1 + \cdots + 0\mathbf{w}_r$; also $\text{span}(S)$ is a subspace (Theorem 4.3.1).

False. When $W(x) \equiv 0$ the Wronskian test is inconclusive (Remark after Fact 4.4.2, item 5).

Problem 2. Multiple choice [4 marks]

Each question is worth 1 mark. Tick exactly one option.

(a) Let $V = \mathbb{R}^2$ with the usual addition $(x_1, x_2) + (y_1, y_2) = (x_1 + y_1,\ x_2 + y_2)$ and the scalar multiplication $k(x_1, x_2) = (kx_1,\ k^2x_2)$. With these operations V is *not* a vector space: exactly one of the following four axioms fails. Which one?

□ A. Axiom 10: $1\mathbf{u} = \mathbf{u}$ □ B. Axiom 9: $k(m\mathbf{u}) = (km)\mathbf{u}$

■ C. Axiom 8: $(k + m)\mathbf{u} = k\mathbf{u} + m\mathbf{u}$ □ D. Axiom 7: $k(\mathbf{u} + \mathbf{v}) = k\mathbf{u} + k\mathbf{v}$

Answer: C (Axiom 8). Take $\mathbf{u} = (1, 1)$, $k = m = 1$: $(k + m)\mathbf{u} = 2(1, 1) = (2, 4)$ but $k\mathbf{u} + m\mathbf{u} = (1, 1) + (1, 1) = (2, 2)$, so Axiom 8 fails. Axiom 10: $1\mathbf{u} = (u_1, u_2)$ holds; Axiom 9: $k(m\mathbf{u}) = (kmu_1, k^2m^2u_2) = (km)\mathbf{u}$ holds; Axiom 7: $k(\mathbf{u}+\mathbf{v}) = (k(u_1+v_1), k^2(u_2+v_2)) = k\mathbf{u}+k\mathbf{v}$ holds.

(b) Which of the following subsets of $V = \mathbb{R}^2$ (with the usual operations) is **not** a subspace of V?

□ A. $W = \{(x, y) : x = 0\}$ (the y-axis) □ B. $W = \{(x, y) : 2x - 5y = 0\}$

□ C. $W = \{(0, 0)\}$ ■ D. $W = \{(x, y) : xy \geq 0\}$

Answer: D. A: subspace (passes the Subspace Test). B: subspace (passes the Subspace Test). C: subspace (passes the Subspace Test). D: $(1, 0),\ (0, -1) \in W$ but their sum $\notin W$.

(c) Let $\mathbf{v}_1 = (1, 0, 0)$ and $\mathbf{v}_2 = (0, 1, 2)$ in $\mathbb{R}^3$. Which of the following vectors belongs to $\text{span}\{\mathbf{v}_1, \mathbf{v}_2\}$?

□ A. $(4, 2, 1)$ □ B. $(1, 2, 3)$ □ C. $(0, 1, 3)$ ■ D. $(4, 1, 2)$

Answer: D. Match $k_1\mathbf{v}_1 + k_2\mathbf{v}_2$ with each candidate (two components fix k_1, k_2; the third must agree). A: $k_1 = 4$, $k_2 = 2$ would force the third component to be $4 \neq 1$. B: $k_1 = 1$, $k_2 = 2$ would force the third component to be $4 \neq 3$. C: $k_1 = 0$, $k_2 = 1$ would force the third component to be $2 \neq 3$. **D**: $k_1 = 4$, $k_2 = 1$ give third component $= 2$ ✓, so $(4, 1, 2) = 4\mathbf{v}_1 + \mathbf{v}_2$.

(d) Let $f_1 = e^{2x}$ and $f_2 = e^{3x}$ be functions in $C^\infty(-\infty, \infty)$. Compute the Wronskian $W(x)$ of f_1, f_2 and choose the correct statement.

□ A. $W(x) = 5e^{5x}$; the set is linearly independent ■ B. $W(x) = e^{5x}$; the set is linearly independent

□ C. $W(x) = e^{5x}$; the set is linearly dependent because $e^{3x} = e^x \cdot e^{2x}$ □ D. $W(x) = e^{6x}$; the set is linearly independent

Answer: B. $W(x) = \begin{vmatrix} e^{2x} & e^{3x} \\ 2e^{2x} & 3e^{3x} \end{vmatrix} = 3e^{5x} - 2e^{5x} = e^{5x} \neq 0$; linearly independent (Fact 4.4.2).

Problem 3. Linear independence and span [4 marks]

Let $S = \{\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3\}$ be the set of vectors in $\mathbb{R}^3$ given by

$$\mathbf{v}_1 = (1, 2, 1), \qquad \mathbf{v}_2 = (2, 1, -1), \qquad \mathbf{v}_3 = (1, 1, 2),$$

and let $\mathbf{u} = (4, 3, -3)$.

(a) [2.5 marks] Determine whether S is linearly independent or linearly dependent. Show all your work.

(b) [1.5 marks] Determine whether $\mathbf{u}$ belongs to $\text{span}(S)$, that is, whether $\mathbf{u}$ is a linear combination of $\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3$. If it is, write $\mathbf{u}$ as such a linear combination; if it is not, explain why. (You may reuse the row operations from part (a).)

Solution.

(a) Set $k_1\mathbf{v}_1 + k_2\mathbf{v}_2 + k_3\mathbf{v}_3 = \mathbf{0}$, i.e. $k_1(1,2,1) + k_2(2,1,-1) + k_3(1,1,2) = (0,0,0)$. Comparing components gives the homogeneous system and its augmented matrix:

$$\begin{cases} k_1 + 2k_2 + k_3 = 0 \\ 2k_1 + k_2 + k_3 = 0 \\ k_1 - k_2 + 2k_3 = 0 \end{cases} \qquad \left[\begin{array}{ccc|c} 1 & 2 & 1 & 0 \\ 2 & 1 & 1 & 0 \\ 1 & -1 & 2 & 0 \end{array}\right] \xrightarrow[-R_1+R_3\to R_3]{-2R_1+R_2\to R_2} \left[\begin{array}{ccc|c} 1 & 2 & 1 & 0 \\ 0 & -3 & -1 & 0 \\ 0 & -3 & 1 & 0 \end{array}\right] \xrightarrow{-R_2+R_3\to R_3} \left[\begin{array}{ccc|c} 1 & 2 & 1 & 0 \\ 0 & -3 & -1 & 0 \\ 0 & 0 & 2 & 0 \end{array}\right]$$

There are three pivots and no free variables, so the homogeneous system has **only the trivial solution** $k_1 = k_2 = k_3 = 0$.

S is linearly independent.

(b) Set $k_1\mathbf{v}_1 + k_2\mathbf{v}_2 + k_3\mathbf{v}_3 = \mathbf{u}$ with $\mathbf{u} = (4, 3, -3)$. The coefficient matrix is the same, so the same row operations apply to the new right-hand side:

$$\left[\begin{array}{ccc|c} 1 & 2 & 1 & 4 \\ 2 & 1 & 1 & 3 \\ 1 & -1 & 2 & -3 \end{array}\right] \xrightarrow[-R_1+R_3\to R_3]{-2R_1+R_2\to R_2} \left[\begin{array}{ccc|c} 1 & 2 & 1 & 4 \\ 0 & -3 & -1 & -5 \\ 0 & -3 & 1 & -7 \end{array}\right] \xrightarrow{-R_2+R_3\to R_3} \left[\begin{array}{ccc|c} 1 & 2 & 1 & 4 \\ 0 & -3 & -1 & -5 \\ 0 & 0 & 2 & -2 \end{array}\right]$$

Back-substitution: $2k_3 = -2 \Rightarrow k_3 = -1$; $-3k_2 - k_3 = -5 \Rightarrow k_2 = 2$; $k_1 + 2k_2 + k_3 = 4 \Rightarrow k_1 = 1$. Hence $\mathbf{u} = \mathbf{v}_1 + 2\mathbf{v}_2 - \mathbf{v}_3$.

$\mathbf{u} \in \text{span}(S)$: $\mathbf{u} = \mathbf{v}_1 + 2\mathbf{v}_2 - \mathbf{v}_3$.

Source: Example 4.4.1 (Linear Dependence in $\mathbb{R}^3$), line 13280; Example 4.2.10 (Vector IS a Linear Combination), line 11774; Example 4.2.11 (Vector Is NOT a Linear Combination), line 11843; Definition 4.4.1 (Linear Independence and Dependence), line 13236; Definition 4.3.1 (Span), line 12614 (`math231_lecture_notes_v02_2026_fully_solved.tex`). New vectors/polynomials; method, structure and hypotheses unchanged.

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Department of Mathematics, Statistics and Physics

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Math 231 – Linear Algebra Quiz 3

Sections 4.1–4.4 (Real Vector Spaces, Subspaces, Spanning Sets, Linear Independence)

Fall 2026 • Thursday, 5 November 2026 • Time: 20 minutes • Total: 10 marks

Name: _____ ID: _____ Section: _____ **Version H**

Tick exactly one box for each True/False statement and exactly one option for each multiple-choice question; an item with more than one answer receives zero. Show all your work in Problem 3. Programmable calculators and mobile phones are not allowed.

Problem 1. True or False [2 marks]

Each statement is worth 0.5 mark. Tick **True** if the statement is always true, otherwise tick **False**. No justification is needed.

	True	False
(a) If $\mathbf{u}$ is a vector in a vector space and k is a scalar with $k\mathbf{u} = \mathbf{0}$, then necessarily $k = 0$.	☐	■
(b) The set $W = \{a + bx + cx^2 : a + b + c = 0\}$ is a subspace of P_2.	■	☐
(c) The set $S = \{(1,-1,0),\ (1,2,1),\ (1,0,1),\ (3,3,-1)\}$ cannot span $\mathbb{R}^3$ because it contains more than three vectors.	☐	■
(d) Every set of five matrices in M_{22} is linearly dependent.	■	☐

False. Theorem 4.1.1(d) says $k = 0$ *or* $\mathbf{u} = \mathbf{0}$. Counterexample: $k = 5$ and $\mathbf{u} = \mathbf{0}$ give $5\mathbf{0} = \mathbf{0}$ with $k \neq 0$ (Theorem 4.1.1(b)).

True. The zero polynomial has $0+0+0 = 0$; the condition is preserved under sums and scalar multiples (Extra Problems 4.2.2).

False. Extra Problems 4.3.2 row reduces to three pivots: the system is consistent for every (a, b, c), so $\text{span}(S) = \mathbb{R}^3$.

True. Fact 4.4.1(3) with $r = 5 > 2 \cdot 2 = 4$.

Problem 2. Multiple choice [4 marks]

Each question is worth 1 mark. Tick exactly one option.

(a) Let $V = \mathbb{R}^3$ with the usual addition $(x_1,x_2,x_3) + (y_1,y_2,y_3) = (x_1 + y_1,\ x_2 + y_2,\ x_3 + y_3)$ and the scalar multiplication $k(x_1,x_2,x_3) = (0,\ kx_2,\ kx_3)$. With these operations V is *not* a vector space: exactly one of the following four axioms fails. Which one?

■ A. Axiom 10: $1\mathbf{u} = \mathbf{u}$

☐ B. Axiom 8: $(k + m)\mathbf{u} = k\mathbf{u} + m\mathbf{u}$

☐ C. Axiom 4: there is a zero vector $\mathbf{0}$ with $\mathbf{0} + \mathbf{u} = \mathbf{u}$

☐ D. Axiom 1: $\mathbf{u} + \mathbf{v}$ is in V (closure under addition)

Answer: A (Axiom 10). Take $\mathbf{u} = (5,1,2)$: $1\mathbf{u} = (0,1,2) \neq \mathbf{u}$, so Axiom 10 fails. Axiom 8: $(k+m)\mathbf{u} = (0,(k+m)u_2,(k+m)u_3) = k\mathbf{u}+m\mathbf{u}$ holds; Axioms 4 and 1 hold for the usual addition.

(b) Which of the following subsets of $V = F(-\infty,\infty)$ (with the usual operations) is **not** a subspace of V?

■ A. $W = \{f : f(0) = 1\}$

☐ B. $W = \{f : f(0) = 0\}$

☐ C. $W = C^1(-\infty,\infty)$, all continuously differentiable functions

☐ D. $W = P_2$, all polynomials of degree at most 2

Answer: A. A: $\mathbf{0} \notin W$. B: subspace (passes the Subspace Test). C: subspace (passes the Subspace Test). D: subspace (passes the Subspace Test).

(c) Let $p_1 = x + x^2$ and $p_2 = 1$ in P_2. Which of the following polynomials belongs to $\text{span}\{p_1, p_2\}$?

■ A. $5 + 2x + 2x^2$ ☐ B. $5 + 2x + 5x^2$ ☐ C. $2 + 5x + 2x^2$ ☐ D. $5 + 2x - 2x^2$

Answer: A. Match $k_1p_1 + k_2p_2$ with each candidate (two coefficients fix k_1, k_2; the third must agree). **A**: $k_1 = 2, k_2 = 5$ give x^2 coefficient $= 2$ ✓, so $5 + 2x + 2x^2 = 2p_1 + 5p_2$. B: $k_1 = 2, k_2 = 5$ would force the x^2 coefficient to be $2 \neq 5$. C: $k_1 = 5, k_2 = 2$ would force the x^2 coefficient to be $5 \neq 2$. D: $k_1 = 2, k_2 = 5$ would force the x^2 coefficient to be $2 \neq -2$.

(d) Let $f_1 = x$ and $f_2 = 1 + x$ be functions in $C^\infty(-\infty,\infty)$. Compute the Wronskian $W(x)$ of f_1, f_2 and choose the correct statement.

☐ A. $W(x) = 1$; the set is linearly independent

☐ B. $W(x) = -1$; the set is linearly dependent because $W(x)$ is negative

■ C. $W(x) = -1$; the set is linearly independent

☐ D. $W(x) = 0$ for all x; the Wronskian test is inconclusive

Answer: C. $W(x) = \begin{vmatrix} x & 1+x \\ 1 & 1 \end{vmatrix} = x - (1+x) = -1 \neq 0$ for every x, so the set is linearly independent (Fact 4.4.2).

Problem 3. Linear independence and span [4 marks]

Let $S = \{p_1, p_2, p_3\}$ be the set of polynomials in P_2 given by

$$p_1 = 2 + x - x^2, \qquad p_2 = 1 + x + x^2, \qquad p_3 = 4 + 3x + x^2,$$

and let $u = 1 + 2x + x^2$.

(a) [2.5 marks] Determine whether S is linearly independent or linearly dependent. Show all your work.

(b) [1.5 marks] Determine whether u belongs to span(S), that is, whether u is a linear combination of p_1, p_2, p_3. If it is, write u as such a linear combination; if it is not, explain why. (You may reuse the row operations from part (a).)

Solution.

(a) Set $k_1p_1 + k_2p_2 + k_3p_3 = 0$, i.e. $k_1(2 + x - x^2) + k_2(1 + x + x^2) + k_3(4 + 3x + x^2) = 0 + 0x + 0x^2$. Comparing coefficients of 1, x and x^2 gives the homogeneous system and its augmented matrix:

$$\begin{cases} 2k_1 + k_2 + 4k_3 = 0 \\ k_1 + k_2 + 3k_3 = 0 \\ -k_1 + k_2 + k_3 = 0 \end{cases} \qquad \left[\begin{array}{ccc|c} 2 & 1 & 4 & 0 \\ 1 & 1 & 3 & 0 \\ -1 & 1 & 1 & 0 \end{array}\right] \xrightarrow{R_1 \leftrightarrow R_2} \left[\begin{array}{ccc|c} 1 & 1 & 3 & 0 \\ 2 & 1 & 4 & 0 \\ -1 & 1 & 1 & 0 \end{array}\right] \xrightarrow[R_1+R_3 \to R_3]{-2R_1+R_2 \to R_2} \left[\begin{array}{ccc|c} 1 & 1 & 3 & 0 \\ 0 & -1 & -2 & 0 \\ 0 & 2 & 4 & 0 \end{array}\right]$$

$$\xrightarrow{2R_2+R_3 \to R_3} \left[\begin{array}{ccc|c} 1 & 1 & 3 & 0 \\ 0 & -1 & -2 & 0 \\ 0 & 0 & 0 & 0 \end{array}\right]$$

The coefficient matrix has only two pivots: the third row is a zero row, so k_3 is a free variable and the homogeneous system has **nontrivial solutions**. For instance $k_3 = 1$ gives $k_2 = -2$ and $k_1 = -1$, so $-p_1 - 2p_2 + p_3 = 0$, i.e. $p_3 = p_1 + 2p_2$.

S is linearly dependent.

(b) Set $k_1p_1 + k_2p_2 + k_3p_3 = u$ with $u = 1 + 2x + x^2$. The coefficient matrix is the same, so the same row operations apply to the new right-hand side:

$$\left[\begin{array}{ccc|c} 2 & 1 & 4 & 1 \\ 1 & 1 & 3 & 2 \\ -1 & 1 & 1 & 1 \end{array}\right] \xrightarrow{R_1 \leftrightarrow R_2} \left[\begin{array}{ccc|c} 1 & 1 & 3 & 2 \\ 2 & 1 & 4 & 1 \\ -1 & 1 & 1 & 1 \end{array}\right] \xrightarrow[R_1+R_3 \to R_3]{-2R_1+R_2 \to R_2} \left[\begin{array}{ccc|c} 1 & 1 & 3 & 2 \\ 0 & -1 & -2 & -3 \\ 0 & 2 & 4 & 3 \end{array}\right]$$

$$\xrightarrow{2R_2+R_3 \to R_3} \left[\begin{array}{ccc|c} 1 & 1 & 3 & 2 \\ 0 & -1 & -2 & -3 \\ 0 & 0 & 0 & -3 \end{array}\right]$$

The last row reads $0k_1 + 0k_2 + 0k_3 = -3$, i.e. $0 = -3$, which is impossible: the system is **inconsistent**. Hence u is not a linear combination of p_1, p_2, p_3.

$u \notin$ span(S).

Source: Example 4.4.2 (Linear Independence in P_3), line 13357; Example 4.2.10 (Vector IS a Linear Combination), line 11774; Example 4.2.11 (Vector Is NOT a Linear Combination), line 11843; Definition 4.4.1 (Linear Independence and Dependence), line 13236; Definition 4.3.1 (Span), line 12614 (`math231_lecture_notes_v02_2026_fully_solved.tex`). New vectors/polynomials; method, structure and hypotheses unchanged.

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Math 231 – Linear Algebra Quiz 3

Sections 4.1–4.4 (Real Vector Spaces, Subspaces, Spanning Sets, Linear Independence)

Fall 2026 • Thursday, 5 November 2026 • Time: 20 minutes • Total: 10 marks

Name: _____ ID: _____ Section: _____ **Version I**

Tick exactly one box for each True/False statement and exactly one option for each multiple-choice question; an item with more than one answer receives zero. Show all your work in Problem 3. Programmable calculators and mobile phones are not allowed.

Problem 1. True or False [2 marks]

Each statement is worth 0.5 mark. Tick **True** if the statement is always true, otherwise tick **False**. No justification is needed.

	True	False
(a) In a vector space V, the equation $k\mathbf{0} = \mathbf{0}$ holds only when the scalar k equals 0.	☐	■
(b) The set $C(-\infty, \infty)$ of all continuous functions on $(-\infty, \infty)$ is a subspace of $F(-\infty, \infty)$.	■	☐
(c) If $\mathbf{u} \in \text{span}\{\mathbf{v}_1, \mathbf{v}_2\}$, then $\mathbf{u} = k_1\mathbf{v}_1 + k_2\mathbf{v}_2$ for some scalars k_1, k_2.	■	☐
(d) Every set of two nonzero vectors in $\mathbb{R}^3$ is linearly independent.	☐	■

False. Theorem 4.1.1(b): $k\mathbf{0} = \mathbf{0}$ for *every* scalar k.

True. The zero function is continuous, and sums and scalar multiples of continuous functions are continuous (Extra Problems 4.2.1).

True. Definition 4.3.1: the span is the set of all linear combinations.

False. Theorem 4.4.1(c): $(1,0,0)$ and $(2,0,0)$ are nonzero but one is a scalar multiple of the other, so the set is linearly dependent.

Problem 2. Multiple choice [4 marks]

Each question is worth 1 mark. Tick exactly one option.

(a) Let $V = \mathbb{R}^2$ with the usual addition $(x_1, x_2) + (y_1, y_2) = (x_1 + y_1,\ x_2 + y_2)$ and the scalar multiplication $k(x_1, x_2) = (x_1,\ kx_2)$. With these operations V is *not* a vector space: exactly one of the following four axioms fails. Which one?

☐ A. Axiom 7: $k(\mathbf{u} + \mathbf{v}) = k\mathbf{u} + k\mathbf{v}$

☐ B. Axiom 9: $k(m\mathbf{u}) = (km)\mathbf{u}$

■ C. Axiom 8: $(k + m)\mathbf{u} = k\mathbf{u} + m\mathbf{u}$

☐ D. Axiom 10: $1\mathbf{u} = \mathbf{u}$

Answer: C (Axiom 8). Take $\mathbf{u} = (5,2)$, $k = 2$, $m = 3$: $(k+m)\mathbf{u} = 5(5,2) = (5,10)$ but $k\mathbf{u} + m\mathbf{u} = (5,4) + (5,6) = (10,10)$, so Axiom 8 fails. Axiom 7: $k(\mathbf{u}+\mathbf{v}) = (u_1+v_1, k(u_2+v_2)) = k\mathbf{u}+k\mathbf{v}$ holds; Axiom 9: $k(m\mathbf{u}) = (u_1, kmu_2) = (km)\mathbf{u}$ holds; Axiom 10: $1\mathbf{u} = \mathbf{u}$ holds.

(b) Which of the following subsets W is a **subspace** of $V = \mathbb{R}^3$ (with the usual operations)?

☐ A. $W = \{(x, y, z) : x \geq 0\}$

☐ B. $W = \{(x, y, z) : x^2 + y^2 + z^2 \leq 1\}$

☐ C. $W = \{(x, y, z) : xyz = 0\}$

■ D. $W = \{(x, y, z) : z = x - 2y\}$

Answer: D. A: $(1,0,0) \in W$ but $-1 \cdot (1,0,0) \notin W$. B: $(1,0,0) \in W$ but $2 \cdot (1,0,0) \notin W$. C: $(1,1,0)$, $(0,0,1) \in W$ but their sum $\notin W$. D: subspace (passes the Subspace Test).

(c) Let $\mathbf{v}_1 = (1,0,1)$ and $\mathbf{v}_2 = (0,1,1)$ in $\mathbb{R}^3$. Which of the following vectors belongs to $\text{span}\{\mathbf{v}_1, \mathbf{v}_2\}$?

☐ A. $(3,-1,4)$ ☐ B. $(3,1,2)$ ■ C. $(3,-1,2)$ ☐ D. $(1,1,1)$

Answer: C. Match $k_1\mathbf{v}_1 + k_2\mathbf{v}_2$ with each candidate (two components fix k_1, k_2; the third must agree). A: $k_1 = 3, k_2 = -1$ would force the third component to be $2 \neq 4$. B: $k_1 = 3, k_2 = 1$ would force the third component to be $4 \neq 2$. **C**: $k_1 = 3, k_2 = -1$ give third component $= 2$ ✓, so $(3,-1,2) = 3\mathbf{v}_1 - \mathbf{v}_2$. D: $k_1 = 1, k_2 = 1$ would force the third component to be $2 \neq 1$.

(d) Let $f_1 = x^2$ and $f_2 = x^3$ be functions in $C^\infty(-\infty, \infty)$. Compute the Wronskian $W(x)$ of f_1, f_2 and choose the correct statement.

☐ A. $W(x) = x^4$; the set is linearly dependent because $W(0) = 0$

■ B. $W(x) = x^4$; the set is linearly independent

☐ C. $W(x) = 5x^4$; the set is linearly independent

☐ D. $W(x) = 0$ for all x; the Wronskian test is inconclusive

Answer: B. $W(x) = \begin{vmatrix} x^2 & x^3 \\ 2x & 3x^2 \end{vmatrix} = 3x^4 - 2x^4 = x^4$. $W(1) = 1 \neq 0$, so $W \not\equiv 0$ and the set is linearly independent (Fact 4.4.2).

Problem 3. Linear independence and span [4 marks]

Let $S = \{\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3\}$ be the set of vectors in $\mathbb{R}^3$ given by

$$\mathbf{v}_1 = (1, -2, 1), \qquad \mathbf{v}_2 = (3, 1, 2), \qquad \mathbf{v}_3 = (5, -3, 4),$$

and let $\mathbf{u} = (2, 1, 3)$.

(a) [2.5 marks] Determine whether S is linearly independent or linearly dependent. Show all your work.

(b) [1.5 marks] Determine whether $\mathbf{u}$ belongs to $\text{span}(S)$, that is, whether $\mathbf{u}$ is a linear combination of $\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3$. If it is, write $\mathbf{u}$ as such a linear combination; if it is not, explain why. (You may reuse the row operations from part (a).)

Solution.

(a) Set $k_1\mathbf{v}_1 + k_2\mathbf{v}_2 + k_3\mathbf{v}_3 = \mathbf{0}$, i.e. $k_1(1,-2,1) + k_2(3,1,2) + k_3(5,-3,4) = (0,0,0)$. Comparing components gives the homogeneous system and its augmented matrix:

$$\begin{cases} k_1 + 3k_2 + 5k_3 = 0 \\ -2k_1 + k_2 - 3k_3 = 0 \\ k_1 + 2k_2 + 4k_3 = 0 \end{cases} \qquad \left[\begin{array}{ccc|c} 1 & 3 & 5 & 0 \\ -2 & 1 & -3 & 0 \\ 1 & 2 & 4 & 0 \end{array}\right] \xrightarrow[-R_1+R_3\to R_3]{2R_1+R_2\to R_2} \left[\begin{array}{ccc|c} 1 & 3 & 5 & 0 \\ 0 & 7 & 7 & 0 \\ 0 & -1 & -1 & 0 \end{array}\right] \xrightarrow{R_2\leftrightarrow R_3} \left[\begin{array}{ccc|c} 1 & 3 & 5 & 0 \\ 0 & -1 & -1 & 0 \\ 0 & 7 & 7 & 0 \end{array}\right]$$

$$\xrightarrow{7R_2+R_3\to R_3} \left[\begin{array}{ccc|c} 1 & 3 & 5 & 0 \\ 0 & -1 & -1 & 0 \\ 0 & 0 & 0 & 0 \end{array}\right]$$

The coefficient matrix has only two pivots: the third row is a zero row, so k_3 is a free variable and the homogeneous system has **nontrivial solutions**. For instance $k_3 = 1$ gives $k_2 = -1$ and $k_1 = -2$, so $-2\mathbf{v}_1 - \mathbf{v}_2 + \mathbf{v}_3 = \mathbf{0}$, i.e. $\mathbf{v}_3 = 2\mathbf{v}_1 + \mathbf{v}_2$.

S is linearly dependent.

(b) Set $k_1\mathbf{v}_1 + k_2\mathbf{v}_2 + k_3\mathbf{v}_3 = \mathbf{u}$ with $\mathbf{u} = (2,1,3)$. The coefficient matrix is the same, so the same row operations apply to the new right-hand side:

$$\left[\begin{array}{ccc|c} 1 & 3 & 5 & 2 \\ -2 & 1 & -3 & 1 \\ 1 & 2 & 4 & 3 \end{array}\right] \xrightarrow[-R_1+R_3\to R_3]{2R_1+R_2\to R_2} \left[\begin{array}{ccc|c} 1 & 3 & 5 & 2 \\ 0 & 7 & 7 & 5 \\ 0 & -1 & -1 & 1 \end{array}\right] \xrightarrow{R_2\leftrightarrow R_3} \left[\begin{array}{ccc|c} 1 & 3 & 5 & 2 \\ 0 & -1 & -1 & 1 \\ 0 & 7 & 7 & 5 \end{array}\right]$$

$$\xrightarrow{7R_2+R_3\to R_3} \left[\begin{array}{ccc|c} 1 & 3 & 5 & 2 \\ 0 & -1 & -1 & 1 \\ 0 & 0 & 0 & 12 \end{array}\right]$$

The last row reads $0k_1 + 0k_2 + 0k_3 = 12$, i.e. $0 = 12$, which is impossible: the system is **inconsistent**. Hence $\mathbf{u}$ is not a linear combination of $\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3$.

$\mathbf{u} \notin \text{span}(S)$.

Source: Example 4.4.1 (Linear Dependence in $\mathbb{R}^3$), line 13280; Example 4.2.10 (Vector IS a Linear Combination), line 11774; Example 4.2.11 (Vector Is NOT a Linear Combination), line 11843; Definition 4.4.1 (Linear Independence and Dependence), line 13236; Definition 4.3.1 (Span), line 12614 (`math231_lecture_notes_v02_2026_fully_solved.tex`). New vectors/polynomials; method, structure and hypotheses unchanged.

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Math 231 – Linear Algebra Quiz 3

Sections 4.1–4.4 (Real Vector Spaces, Subspaces, Spanning Sets, Linear Independence)

Fall 2026 • Thursday, 5 November 2026 • Time: 20 minutes • Total: 10 marks

Name: _____ ID: _____ Section: _____ **Version J**

Tick exactly one box for each True/False statement and exactly one option for each multiple-choice question; an item with more than one answer receives zero. Show all your work in Problem 3. Programmable calculators and mobile phones are not allowed.

Problem 1. True or False [2 marks]

Each statement is worth 0.5 mark. Tick **True** if the statement is always true, otherwise tick **False**. No justification is needed.

	True	False
(a) In a real vector space, the scalars may be any real numbers, including irrational numbers such as $\sqrt{2}$.	■	□
True. Scalars are real numbers (Definition 4.1.1); the notes use $k = \sqrt{2}$ in Extra Problems 4.2.9(c).		
(b) If A is an $m \times n$ matrix and $\mathbf{b} \neq \mathbf{0}$, then the solution set of $A\mathbf{x} = \mathbf{b}$ is a subspace of $\mathbb{R}^n$.	□	■
False. $A\mathbf{0} = \mathbf{0} \neq \mathbf{b}$, so the zero vector is not a solution; condition (a) fails (Example 4.2.8).		
(c) The polynomial x^2 belongs to $\text{span}\{1 + x^2,\ x\}$.	□	■
False. $k_1(1 + x^2) + k_2 x = k_1 + k_2 x + k_1 x^2$; matching x^2 requires $k_1 = 0$ (constant term) and $k_1 = 1$ (x^2 term), impossible.		
(d) A set consisting of a single nonzero vector is linearly independent.	■	□
True. Theorem 4.4.1(b).		

Problem 2. Multiple choice [4 marks]

Each question is worth 1 mark. Tick exactly one option.

(a) Let $V = \mathbb{R}^3$ with the usual addition $(x_1, x_2, x_3) + (y_1, y_2, y_3) = (x_1 + y_1,\ x_2 + y_2,\ x_3 + y_3)$ and the scalar multiplication $k(x_1, x_2, x_3) = (x_1,\ x_2,\ kx_3)$. With these operations V is *not* a vector space: exactly one of the following four axioms fails. Which one?

□ A. Axiom 6: $k\mathbf{u}$ is in V (closure under scalar multiplication)

■ B. Axiom 8: $(k + m)\mathbf{u} = k\mathbf{u} + m\mathbf{u}$

□ C. Axiom 10: $1\mathbf{u} = \mathbf{u}$

□ D. Axiom 9: $k(m\mathbf{u}) = (km)\mathbf{u}$

Answer: B (Axiom 8). Take $\mathbf{u} = (1, 1, 1)$, $k = m = 1$: $(k + m)\mathbf{u} = 2\mathbf{u} = (1, 1, 2)$ but $k\mathbf{u} + m\mathbf{u} = (1, 1, 1) + (1, 1, 1) = (2, 2, 2)$, so Axiom 8 fails. Axiom 6 holds since $(u_1, u_2, ku_3) \in \mathbb{R}^3$; Axiom 10: $1\mathbf{u} = \mathbf{u}$ holds; Axiom 9: $k(m\mathbf{u}) = (u_1, u_2, kmu_3) = (km)\mathbf{u}$ holds.

(b) Which of the following subsets of $V = P_2$ (with the usual operations) is **not** a subspace of V?

□ A. $W = \{p \in P_2 : p(0) = 0\}$

■ B. $W = \{a + bx + cx^2 : a + b + c = 2\}$

□ C. $W = \{a + bx + cx^2 : a = 2c\}$

□ D. $W = \{p \in P_2 : p(1) = 0\}$

Answer: B. A: subspace (passes the Subspace Test). B: $\mathbf{0} \notin W$. C: subspace (passes the Subspace Test). D: subspace (passes the Subspace Test).

(c) Let $p_1 = 1 + 2x$ and $p_2 = x^2$ in P_2. Which of the following polynomials belongs to $\text{span}\{p_1, p_2\}$?

□ A. $2 + 2x - x^2$ ■ B. $2 + 4x - x^2$ □ C. $4 + 2x - x^2$ □ D. $1 + 4x - x^2$

Answer: B. Match $k_1 p_1 + k_2 p_2$ with each candidate (two coefficients fix k_1, k_2; the third must agree). A: $k_1 = 2, k_2 = -1$ would force the x coefficient to be $4 \neq 2$. **B**: $k_1 = 2, k_2 = -1$ give x coefficient $= 4$ ✓, so $2 + 4x - x^2 = 2p_1 - p_2$. C: $k_1 = 4, k_2 = -1$ would force the x coefficient to be $8 \neq 2$. D: $k_1 = 1, k_2 = -1$ would force the x coefficient to be $2 \neq 4$.

(d) Let $f_1 = e^{-x}$ and $f_2 = e^{x}$ be functions in $C^{\infty}(-\infty, \infty)$. Compute the Wronskian $W(x)$ of f_1, f_2 and choose the correct statement.

■ A. $W(x) = 2$; the set is linearly independent

□ B. $W(x) = 0$ for all x; the Wronskian test is inconclusive

□ C. $W(x) = -2$; the set is linearly independent

□ D. $W(x) = 2$; the set is linearly dependent because $W(x)$ is constant

Answer: A. $W(x) = \begin{vmatrix} e^{-x} & e^{x} \\ -e^{-x} & e^{x} \end{vmatrix} = e^{-x}e^{x} - e^{x}(-e^{-x}) = 1 + 1 = 2 \neq 0$, so the set is linearly independent (Fact 4.4.2).

Problem 3. Linear independence and span [4 marks]

Let $S = \{p_1, p_2, p_3\}$ be the set of polynomials in P_2 given by

$$p_1 = 1 + 2x, \qquad p_2 = x - x^2, \qquad p_3 = 1 + x^2,$$

and let $u = 1 - x + x^2$.

(a) [2.5 marks] Determine whether S is linearly independent or linearly dependent. Show all your work.

(b) [1.5 marks] Determine whether u belongs to span(S), that is, whether u is a linear combination of p_1, p_2, p_3. If it is, write u as such a linear combination; if it is not, explain why. (You may reuse the row operations from part (a).)

Solution.

(a) Set $k_1p_1 + k_2p_2 + k_3p_3 = 0$, i.e. $k_1(1 + 2x) + k_2(x - x^2) + k_3(1 + x^2) = 0 + 0x + 0x^2$. Comparing coefficients of 1, x and x^2 gives the homogeneous system and its augmented matrix:

$$\begin{cases} k_1 + k_3 = 0 \\ 2k_1 + k_2 = 0 \\ -k_2 + k_3 = 0 \end{cases} \qquad \left[\begin{array}{ccc|c} 1 & 0 & 1 & 0 \\ 2 & 1 & 0 & 0 \\ 0 & -1 & 1 & 0 \end{array}\right] \xrightarrow{-2R_1+R_2\to R_2} \left[\begin{array}{ccc|c} 1 & 0 & 1 & 0 \\ 0 & 1 & -2 & 0 \\ 0 & -1 & 1 & 0 \end{array}\right] \xrightarrow{R_2+R_3\to R_3} \left[\begin{array}{ccc|c} 1 & 0 & 1 & 0 \\ 0 & 1 & -2 & 0 \\ 0 & 0 & -1 & 0 \end{array}\right]$$

There are three pivots and no free variables, so the homogeneous system has **only the trivial solution** $k_1 = k_2 = k_3 = 0$.

S is linearly independent.

(b) Set $k_1p_1 + k_2p_2 + k_3p_3 = u$ with $u = 1 - x + x^2$. The coefficient matrix is the same, so the same row operations apply to the new right-hand side:

$$\left[\begin{array}{ccc|c} 1 & 0 & 1 & 1 \\ 2 & 1 & 0 & -1 \\ 0 & -1 & 1 & 1 \end{array}\right] \xrightarrow{-2R_1+R_2\to R_2} \left[\begin{array}{ccc|c} 1 & 0 & 1 & 1 \\ 0 & 1 & -2 & -3 \\ 0 & -1 & 1 & 1 \end{array}\right] \xrightarrow{R_2+R_3\to R_3} \left[\begin{array}{ccc|c} 1 & 0 & 1 & 1 \\ 0 & 1 & -2 & -3 \\ 0 & 0 & -1 & -2 \end{array}\right]$$

Back-substitution: $-k_3 = -2 \Rightarrow k_3 = 2$; $k_2 - 2k_3 = -3 \Rightarrow k_2 = 1$; $k_1 + k_3 = 1 \Rightarrow k_1 = -1$. Hence $u = -p_1 + p_2 + 2p_3$.

$u \in \text{span}(S)$: $u = -p_1 + p_2 + 2p_3$.

Source: Example 4.4.2 (Linear Independence in P_3), line 13357; Example 4.2.10 (Vector IS a Linear Combination), line 11774; Example 4.2.11 (Vector Is NOT a Linear Combination), line 11843; Definition 4.4.1 (Linear Independence and Dependence), line 13236; Definition 4.3.1 (Span), line 12614 (`math231_lecture_notes_v02_2026_fully_solved.tex`). New vectors/polynomials; method, structure and hypotheses unchanged.